

Team Name

CODESLAYER 2K26

DevSphereIndia



Project Title : **RepairOS** *Diagnose Before You Replace*

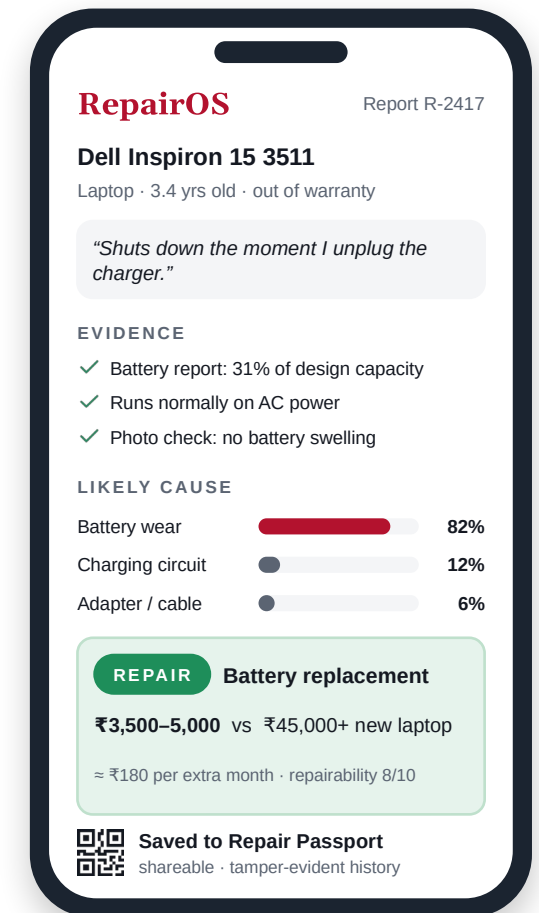
An evidence-first AI platform that tells phone and laptop owners what failed, what a fair fix should cost, and whether to repair, service or replace.

Team Name : Team Name

Team Members *(Names + Roles Briefly):*

- **Member 1** — Team lead · product, decision logic & pitch
- **Member 2** — Frontend · Next.js PWA, self-tests, Passport UI
- **Member 3** — Backend · FastAPI, Bayesian engine, data model
- **Member 4** — AI/ML · LLM extraction, OCR/vision, RAG

Tracks Chosen: **Open Innovation** with an AI/ML core and a Sustainability outcome



Illustrative prototype screen

PROBLEM STATEMENT & TARGET AUDIENCE



DevSphereIndia



Team Name

When a phone or laptop fails, owners **decide blind** — so fixable devices get replaced, and repairs get overpaid.



Problem We're Solving

A device fault is an information problem before it is a technical one. Owners can't tell what failed, what a fair repair should cost, or whether the device is still worth fixing — and the people who can tell them usually profit from the answer.

43%

of Indian smartphone users who used a service centre found repair cost high to very high¹

4 in 10

needed more than one service-centre visit to get a single issue resolved¹

1.75 Mt

e-waste generated in India in FY24 — up 72.5% in five years²

72%

of a smartphone's climate impact comes from making, shipping and disposing of it³

WHERE THE REPAIR DECISION BREAKS TODAY

01

Device fails



02

Owner can't pin down the fault



03

Opaque quote, often a full-module swap



04

Replaced “to be safe”



05

Avoidable cost and e-waste



Target Users

B2C · PRIMARY

- Phone & laptop owners aged 18–40: students, young professionals, gig workers
- Out-of-warranty devices 2–5 years old — where repair-or-replace is decided
- Used-device buyers and sellers who need proof of condition

B2B · REVENUE

- Independent repair shops: pre-triaged leads, digital job cards
- Refurbishers & device insurers: grading and claim triage
- Campus and SME IT teams: fleet repair-vs-replace calls

740M+

active smartphones in India — the B2C base, before laptops⁴

¹ Counterpoint Research, India Smartphone After-Sales Service Consumer Study (Sep 2025) ² CPCB data, MoHUA written reply in Rajya Sabha (16 Dec 2024)

³ European Environmental Bureau / Coolproducts, “Coolproducts don't cost the Earth” (2019) ⁴ Counterpoint Research, active smartphone installed base (2025)



DevSphereIndia



OUR UNIQUE SOLUTION

Team Name

RepairOS turns a vague complaint into **device evidence, a ranked diagnosis and a costed decision** — in under five minutes.

⚡ Key Features



Identify the device

Snap the model label or service tag; OCR pulls specs, device age and known failure patterns for that model.



Collect real evidence

Adaptive English/Hinglish questions, photos, in-browser self-tests (touch, audio, camera, sensors) and OS health reports such as battery wear.



Diagnose with confidence

A Bayesian engine ranks likely faults with a confidence score and asks the single most informative next question.



Decide in rupees

Repairability score on the six factors used by India's Repairability Index, plus ₹ per extra month of life, fair-price band and quote check.



Track in a Repair Passport

A shareable, tamper-evident device history (QR) for resale, warranty claims and the next diagnosis.



Why It's Different

	AI chatbots	DIY repair guides	Repair / trade-in apps	OEM apps	RepairOS
Evidence from the device itself	☐	☐	☐	☒	☒
Ranked diagnosis + confidence	☒	☒	☒	☒	☒
Repair-vs-replace economics (₹)	☒	☐	☒	☐	☒
Outcome-neutral advice	☒	☒	☐	☐	☒
Persistent device history	☐	☐	☒	☒	☒

☒ Full ☒ Partial ☐ None

NOVELTY FACTOR

Evidence-first: reads the device's own health data, not just the user's words.

Decision-first: answers “what should I do?” in rupees and months — then remembers it.

Comparison uses representative products and their publicly documented features: AI chatbots (ChatGPT, Gemini), DIY guides (iFixit), repair/trade-in apps (Cashify), OEM diagnostics (Samsung Members).

TECH STACK + ARCHITECTURE



DevSphereIndia



Team Name

Language models read the messy input; **a deterministic, auditable engine makes the call.**



Tech Stack

FRONTEND

Next.js 14 · React · TypeScript · Tailwind · PWA with Web APIs for self-tests

BACKEND

Python · FastAPI · Pydantic JSON contracts

DIAGNOSTIC ENGINE

Bayesian fault model (NumPy) · YAML rule graph · info-gain question picker

AI SERVICES

LLM API in JSON-schema mode · vision model · Tesseract OCR

DATA

Supabase: PostgreSQL, Auth, Storage · pgvector for RAG

OUTPUT

WeasyPrint PDF · QR share links · SHA-256 hash-chained Passport

DEPLOY

Vercel · Render · GitHub Actions CI



Architecture

Client · Next.js PWA

Guided intake · self-tests · uploads · Passport viewer

API · FastAPI orchestrator

Sessions · auth · schema-validated JSON contracts

REPAIROS CORE — every verdict is traceable to its evidence

1 Evidence extractor

LLM → JSON symptoms, OCR, vision cues, log parsers

2 Diagnostic engine

Bayesian ranking + next-best question

3 Safety gate

Swelling, liquid or heat → stop & route to a pro

4 Decision engine

Repairability, ₹ economics, fair-price check

5 Report & Passport

PDF report, QR link, hash-chained history

AI services

LLM · vision · OCR APIs

Device knowledge base

Specs · failure priors · parts · ₹ bands (pgvector)

Supabase

Postgres · Auth · Storage



Learning loop: shop-verified repair outcomes update each model's failure priors — every repair sharpens the next diagnosis.



DevSphereIndia



FEASIBILITY AND SHOWSTOPPERS

Team Name

Scoped to **one believable end-to-end flow** —
built on APIs, rules and public data, with no model training required.



Feasibility

2

device categories:
phones + laptops

10

high-frequency fault
families

25

top-selling models
seeded in India

- No model training: pretrained LLM/vision APIs; reasoning is rules + Bayes in plain Python
- Real device evidence is free: Windows battery report, macOS system_profiler, browser sensor APIs
- Knowledge base seeded from public spec sheets, service manuals and parts listings
- Skills map 1:1 to modules (frontend, engine, AI, product); free-tier hosting



Showstoppers

RISK

MITIGATION

- | | |
|--|--|
| • Wrong or over-confident diagnosis | LLM only extracts; the Bayesian engine decides and shows confidence. Low confidence $\Rightarrow$ "get it inspected", never "replace". |
| • Unsafe DIY advice | Hard safety gate: swollen battery, liquid or burning smell stops self-help and routes to a professional. |
| • Thin or volatile price data | ₹ bands labelled indicative, with source and date; partner-shop quotes refine them. |
| • Long tail of device models | Category-level priors as fallback; manual model entry when OCR fails. |
| • LLM latency, cost or outage | Schema-bound, cached calls; rule-only fallback keeps the core flow running. |
| • Privacy of IMEI and photos | Hashed device IDs, consent-first uploads; photos deleted unless saved to the Passport. |

24-HOUR BUILD PLAN · demo-first: one complete flow ("laptop won't hold charge") working by hour 15

0–4h

Scaffold: UI, DB, API contracts

4–10h

Knowledge base + Bayesian engine

10–15h

AI evidence: LLM, OCR, battery report

15–19h

Decision, economics, safety gate

19–22h

Report + Repair
Passport

22–24h

Test & demo
polish

USP(UNIQUE SELLING PRICE) & BUSINESS MODEL



DevSphereIndia



Team Name

We earn whether a device is **repaired, resold or recycled** — so the advice stays honest.

☆ USP (Unique Selling Proposition)

- 01

Evidence, not opinions
Reads the device's own health data — battery wear, self-tests, photos — and shows the confidence behind every verdict.
- 02

The decision, not just a diagnosis
Repair vs replace in ₹ per month of extra life, with a fair-price band and a check on any quote a shop gives you.
- 03

Outcome-neutral by design
Revenue from every path — repair, resale, recycling — so there is no reason to push a new device.
- 04

Repair Passport
A portable, tamper-evident service history that makes used devices easier to trust, insure and resell.

WHY NOW India's Repairability Index framework (DoCA, 2025) and the EU's A–E repairability label (June 2025) score the same six factors RepairOS uses. We are the consumer-facing layer for them.

📈 Business Model

REVENUE STREAM	WHO PAYS	PRICING (PLANNED)
Free diagnosis + basic Passport	Consumers	₹0 • acquisition
Verified Repair Report for resale or insurance	Consumers	₹99 / report
Booked-repair commission	Partner shops	8–10% of ticket
Shop OS: pre-triage, job cards, parts lookup	Repair shops	₹499 / month
Diagnostics API: claim & grading triage	Insurers, refurbishers, IT fleets	Per device
Trade-in & recycling referrals	Refurbishers, EPR recyclers	Per referral

GO-TO-MARKET

Pilot • 0–6 months

One metro repair hub + 2 campuses; 30 partner shops



Scale • 6–18 months

Multi-city shop network; verified outcomes sharpen priors



Platform • 18 months+

B2B API for insurers, refurbishers and OEM after-sales